Hui Lin

Machine Learning Researcher & Engineer · Foundation Models · Large-Scale ML · Multimodal AI

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Professional Summary

Machine Learning Researcher and Engineer with a Ph.D. in Electrical and Computer Engineering and experience spanning **foundation models, multimodal AI, recommendation systems, computer vision, and large-scale machine learning**. Experienced in developing Transformer and representation-learning models, building ML pipelines on millions of real-world samples, and translating research into practical systems through rigorous evaluation and cross-functional collaboration. Published **14 first-author papers** with **1,277+ citations**.

Skills

Machine Learning: Transformers, ResNet, self-supervised learning, contrastive learning, representation learning, classification, regression, domain adaptation

Foundation & Multimodal AI: LLM, VLMs, QLoRA, multimodal learning, computer vision, physiological time series, linear probing, constrained decoding

Large-Scale ML: recommendation systems, ranking/prediction, feature engineering, offline evaluation, monitoring, A/B testing, cross-dataset validation

Programming & Systems: Python, C++, SQL, PyTorch, TensorFlow, Spark/Hive, Linux, Docker, Git

Experience

Senior Research Scientist 02.2025 – Present
OPPO US Research Center *Palo Alto, CA*

- Developed scalable **Spark/Hive ML pipelines** for a recommendation system serving **8M+ users**, supporting feature engineering, offline evaluation, production monitoring, and A/B testing.
- Developed a **self-supervised foundation model** to learn generalizable representations from large-scale physiological time-series data; evaluated pretrained features across multiple downstream classification and regression tasks.
- Developed compact **ResNet/Transformer** models using **7M+ measurements from 3,000+ subjects** for large-scale real-world prediction tasks.
- Improved model generalization across **multiple datasets and devices** using contrastive learning, calibration, subject-level validation, and systematic failure analysis.
- Built reproducible end-to-end workflows for **data preprocessing, model training, benchmarking, and evaluation**, and collaborated with research, engineering, and product teams on model integration and deployment.

Graduate Researcher 03.2020 – 09.2025
Northwestern University *Evanston, IL*

- Led machine-learning research spanning **computer vision, multimodal learning, representation learning, domain adaptation, experimental design, and quantitative evaluation**.
- Developed GAN-based **unsupervised domain adaptation** for cross-modality medical image segmentation, improving Dice by **11.4%** and ranking **5th globally** in MICCAI FLARE.
- Developed Transformer, CNN, detection, and segmentation models for visual prediction, achieving **85.70% sensitivity at 95% specificity**; ranked **5th** in ISBI JustRAIGS and **3rd** in MICCAI ARCADE.

Multimodal Foundation Model Research 2026
MICCAI FLARE 2026

- Adapted **InternVL3-8B** across multiple medical-image tasks using task-specific **QLoRA adapters**, ROI-aware preprocessing, constrained decoding, and calibration; built a reproducible **Docker pipeline** for training, inference, and evaluation.

Research Assistant 2015 – 2019
Huazhong University of Science and Technology *Wuhan, China*

- Developed CNN and CycleGAN methods for classification, weakly supervised localization, and synthetic data generation; related publications have received **690+ citations**.

Education

Ph.D. Electrical and Computer Engineering, GPA 3.9/4.0, Northwestern University 09.2019 – 09.2025
M.S. Mechanical Engineering, Rank 1, Huazhong University of Science and Technology 09.2016 – 06.2019
B.S. Materials Processing and Control Engineering, Rank 3, HUST Qiming College 09.2012 – 06.2016

Selected Publications

- **Longitudinal Wrist PPG Analysis for Reliable Hypertension Risk Screening Using Deep Learning** | ICASSP 2025 | **Lin, H.**, et al.
- **DRL-STNet: Unsupervised Domain Adaptation for Cross-Modality Medical Image Segmentation** | MICCAI 2024 | **Lin, H.**, et al.
- **GyroFormer: 3D Rotational BCG for Hypertension Screening** | EMBC 2026 | **Lin, H.**, et al.